

Task 1: Loan Approval Prediction Using Machine Learning

GitHub Repository

Repository Link: <https://github.com/Harshi1904/Loan-Approval-Prediction>

Objective

The objective of this project is to build a supervised machine learning model that predicts whether a loan application will be approved based on applicant information such as income, credit history, education, employment status, and property area.

Dataset

Loan Approval Prediction Dataset (Kaggle)

Methodology

Data Preprocessing

- Loaded and explored the dataset.
- Handled missing values using mode and median imputation.
- Encoded categorical variables using Label Encoding.
- Scaled numerical features using StandardScaler.

Class Imbalance Handling

The dataset contained more approved loans than rejected loans. SMOTE (Synthetic Minority Oversampling Technique) was applied to balance the classes before model training.

Models Evaluated

1. Logistic Regression
2. Random Forest

Evaluation Metrics

The models were evaluated using:

- Precision
- Recall
- F1 Score
- ROC-AUC Score

Results

Model	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.849	0.929	0.888	0.833
Random Forest	0.843	0.882	0.862	0.772

Best Model

Logistic Regression achieved the highest ROC-AUC score (0.833) and provided the best overall performance.

Business Interpretation

- Credit History was one of the most influential factors affecting loan approval.
- Applicants with stronger financial backgrounds had a higher probability of approval.
- The model can assist financial institutions in making faster and more consistent lending decisions.

Deployment Recommendation

A decision threshold of 0.6 is recommended to reduce risky approvals while maintaining an acceptable approval rate.

Screenshots Included

1. Dataset overview
2. Missing value analysis
3. Model comparison table
4. Feature importance chart
5. ROC-AUC results

Conclusion

A machine learning pipeline was successfully developed for loan approval prediction. After preprocessing, scaling, and balancing the data using SMOTE, Logistic Regression emerged as the best-performing model. The solution can support automated loan screening and improve decision-making efficiency.

 `import pandas as pd`

```
results = pd.DataFrame({
    'Model': ['Logistic Regression', 'Random Forest'],
    'Precision': [lr_precision, rf_precision],
    'Recall': [lr_recall, rf_recall],
    'F1 Score': [lr_f1, rf_f1],
    'ROC-AUC': [lr_auc, rf_auc]
})
```

results

...

	Model	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.849462	0.929412	0.887640	0.833437
1	Random Forest	0.842697	0.882353	0.862069	0.772446



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	Feature	Importance
9	Credit_History	0.211463
5	ApplicantIncome	0.203528
7	LoanAmount	0.175169
6	CoapplicantIncome	0.106663
10	Property_Area	0.084558
2	Dependents	0.075341
8	Loan_Amount_Term	0.038593
1	Married	0.036788
3	Education	0.028628
4	Self_Employed	0.019783



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